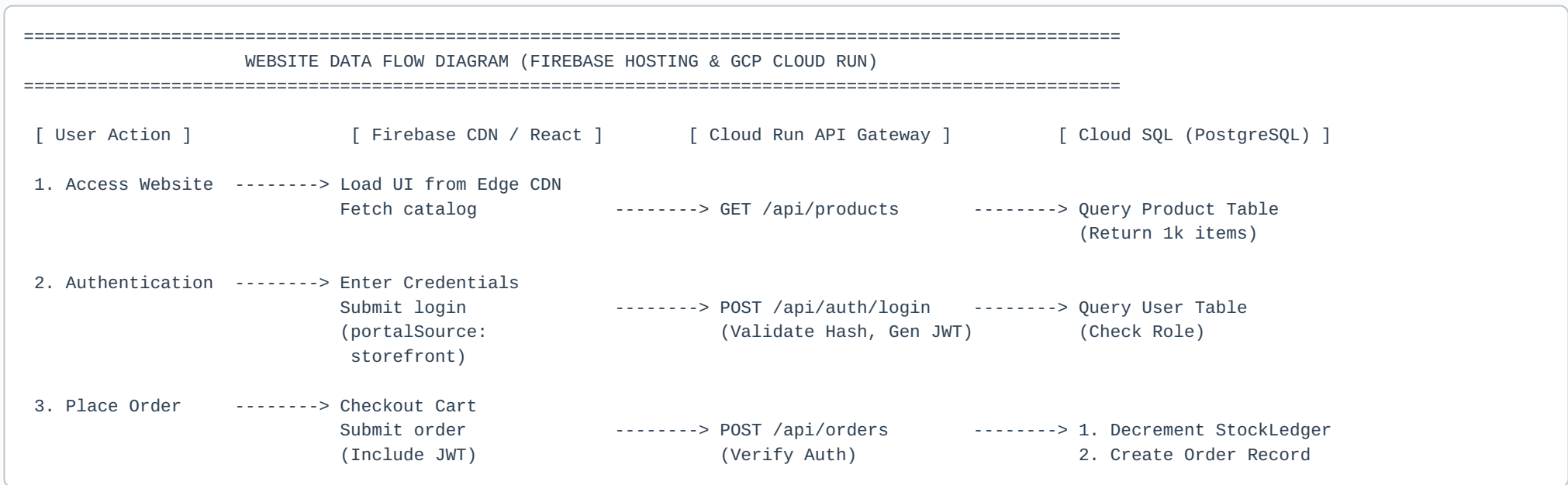


Sapna Sarees - Comprehensive Architecture & Data Flow (Google Cloud Platform)

1. Website-Specific Data Flow Diagram (Firebase Hosting + Cloud Run)

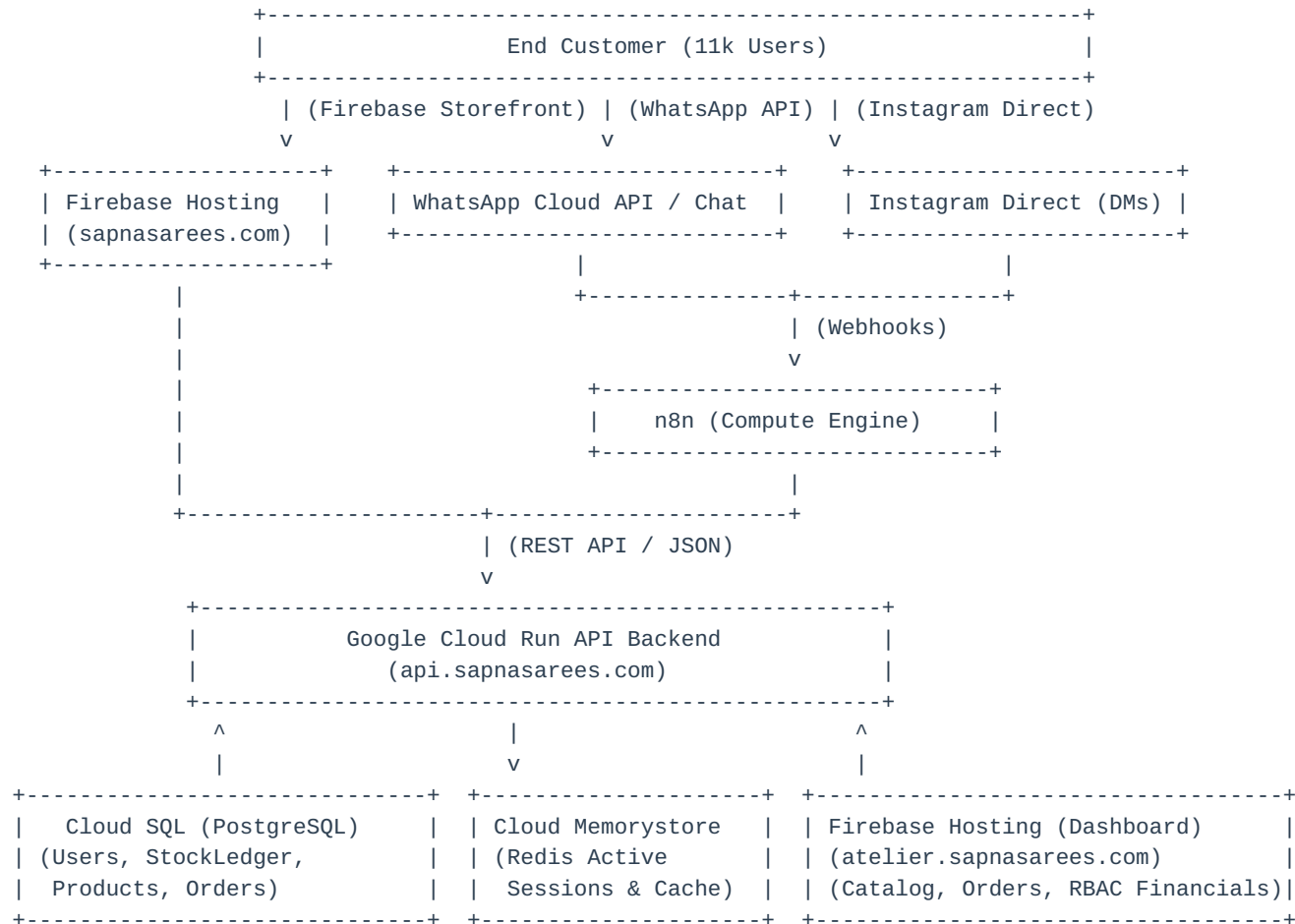
The web storefront routes interactions through HTTPS/REST using React Axios calls via Firebase Hosting to Cloud Run. Cloud SQL maintains atomic database operations across the storefront, ensuring data integrity during catalog browsing and checkout.



2. Comprehensive Data Flow Diagram (DFD)

The platform routes all customer interactions through Google Cloud Run, maintaining atomic database operations in Cloud SQL across the web storefront, WhatsApp, and Instagram.

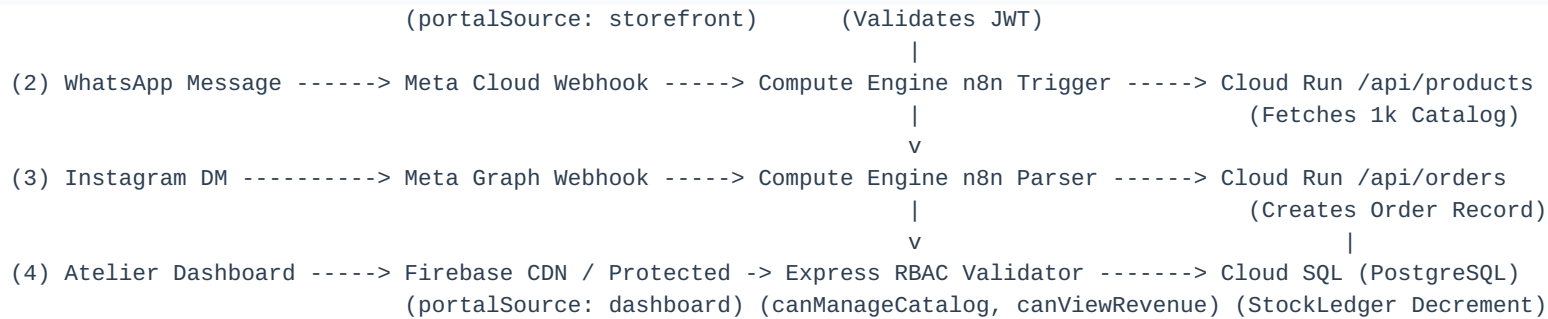
LEVEL 0: CONTEXT DIAGRAM (GCP ARCHITECTURE)



LEVEL 1: DETAILED DATA PIPELINE

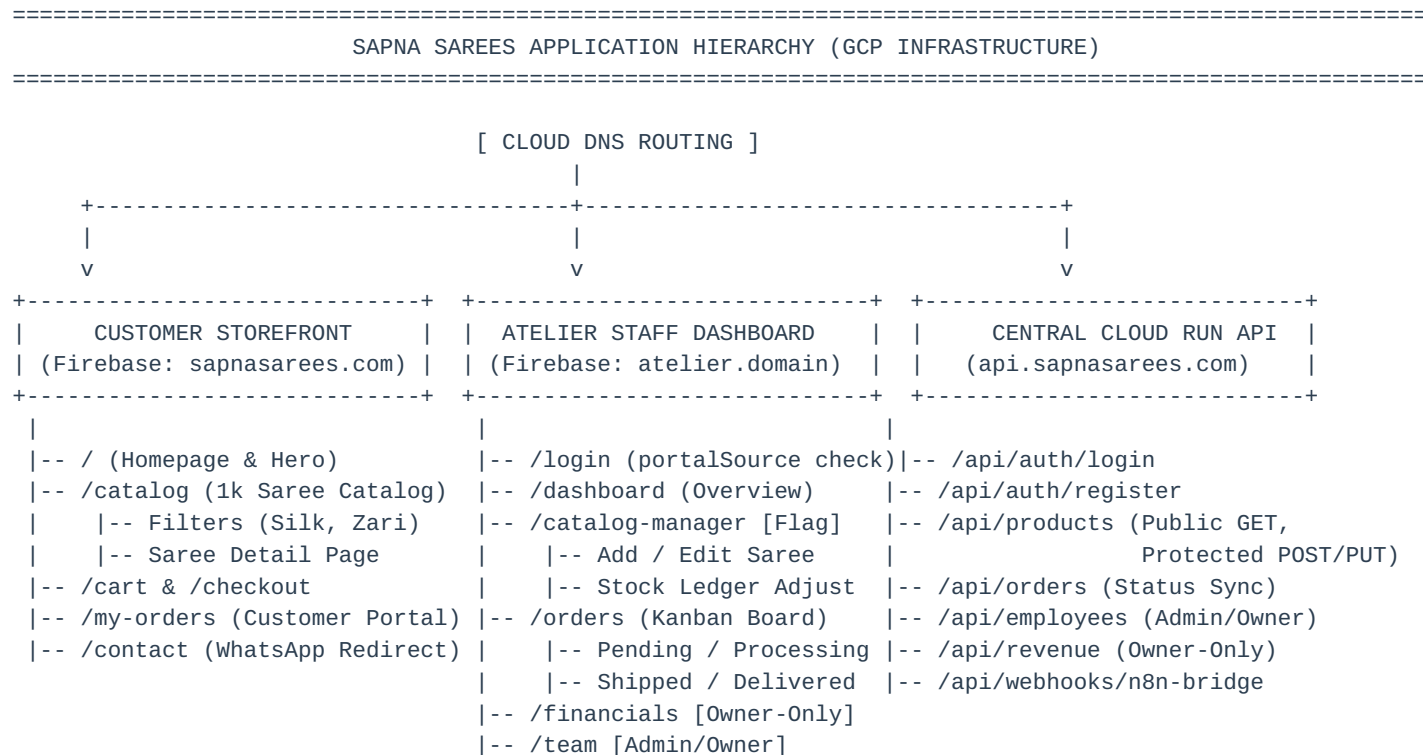
[Customer Touchpoint] [Ingestion Layer] [GCP Core Processing] [Persistence Layer]

(1) Web Storefront -----> Firebase CDN / HTTPS ---> Cloud Run Express Middleware ---> Cloud SQL (User Table)



3. Website & Application Structural Diagram

The architecture separates the customer-facing storefront from the staff management portal on Firebase Hosting, routing through Cloud Run.

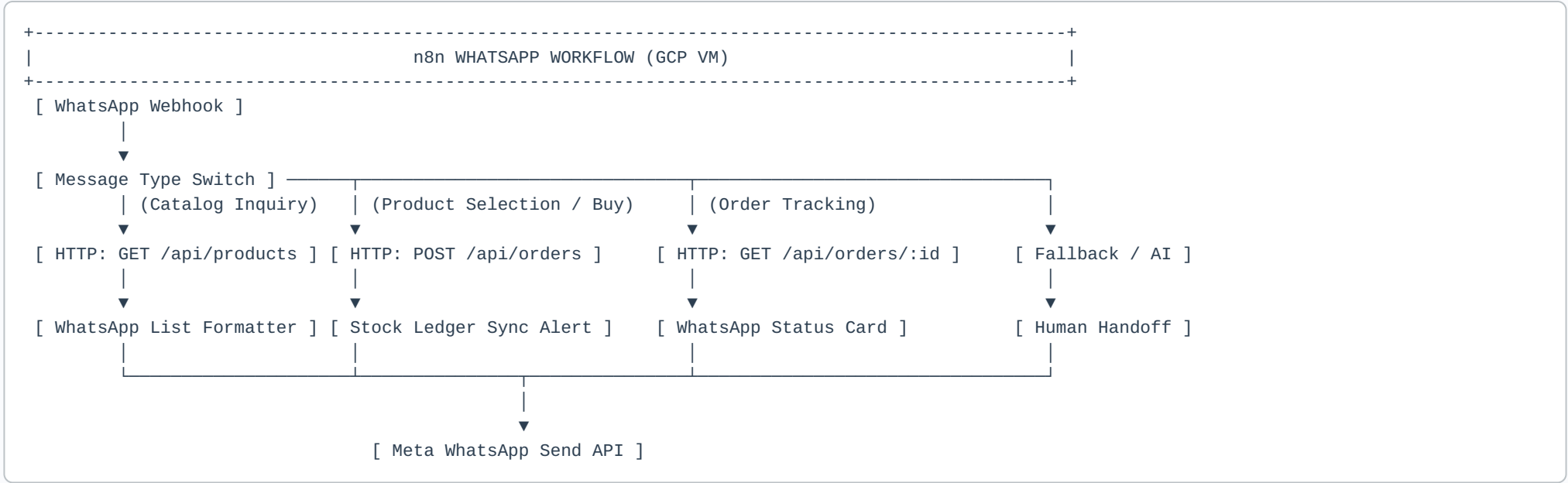


4. n8n Automation Workflow: WhatsApp & Instagram (Hosted on Compute Engine)

n8n on Google Compute Engine serves as the orchestrator between Meta's APIs and the Cloud Run backend service.

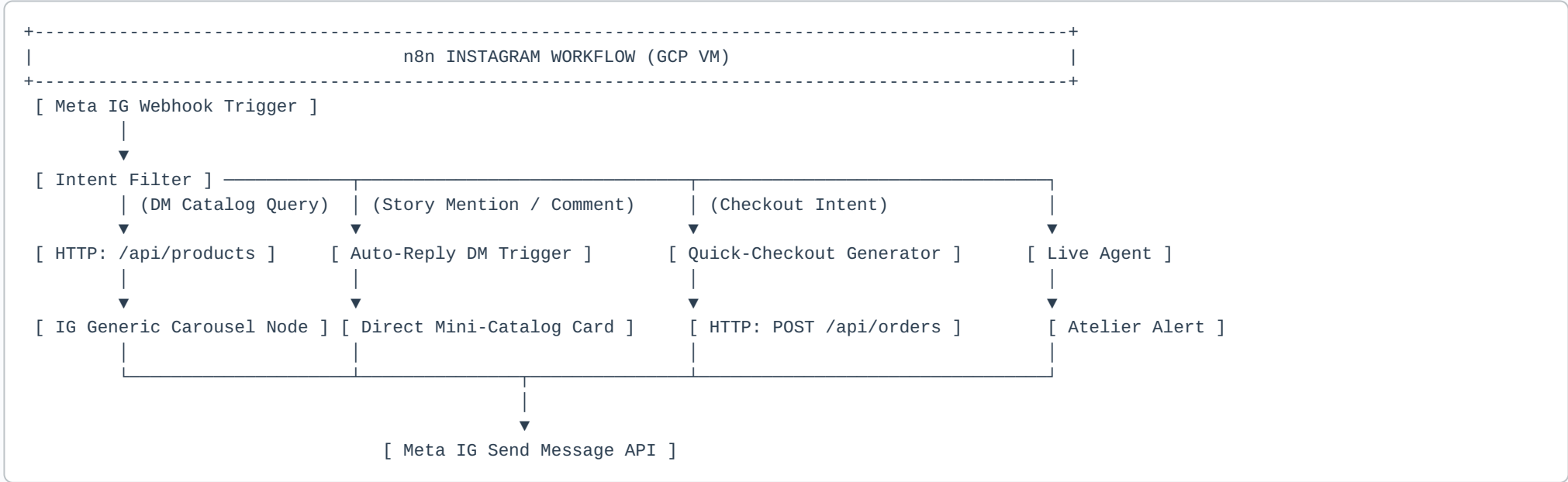
4.1 n8n WhatsApp Flow Architecture

- **Webhook Listener Node:** Receives incoming messages and button clicks from the WhatsApp Business Cloud API.
- **Conditional Router Node (Switch):** Evaluates payload type (Text Keywords, Interactive Button Reply, Order Status Check).
- **HTTP Request Node (API Fetch):** Queries Cloud Run API (GET /api/products) to retrieve real-time inventory from Cloud SQL.
- **Format & Template Node:** Formats the inventory data into Meta-compliant Multi-Product Messages (MPM).
- **Order Generation & Sync Node:** Dispatches a POST payload to Cloud Run to decrement stock, create an order, and update the Kanban board.



4.2 n8n Instagram Automation Flow Architecture

- **Instagram Webhook Trigger:** Catches incoming Direct Messages (DMs), Story Mentions, and Post Comments.
- **Intent Classification Node:** Routes queries to automated mini-catalogs or Cloud Run backend product searches.
- **Instagram Carousel Node:** Generates an Instagram Generic Template Carousel (image cards, price, fabric details, "Buy on Web" CTA).
- **Unified Order Bridge:** Captures shipping info and generates a pre-filled checkout link or updates Cloud SQL directly.



5. Consolidated Synchronization Summary

Channel	Trigger Mechanism	GCP / n8n Processing Role	Database Action (Cloud SQL PostgreSQL)
Website	Firebase Hosting / Axios Calls	Direct Cloud Run REST API Ingestion	Immediate CRUD on Product, Order, StockLedger
WhatsApp	Meta Cloud API Webhook	Compute Engine n8n parses commands & formats lists	Decrements StockLedger, writes new Order
Instagram	Meta Graph API Webhook	Compute Engine n8n automates DMs & carousel cards	Queries Product catalog, records incoming DM orders
Atelier Staff	Firebase Staff Dashboard	Direct Cloud Run REST API with RBAC verification	Real-time order status updates, catalog editing, audit trails